

Geodesics of Attention

Cinema, Gesture, and the Perceptual Manifold

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Abstract

We propose a unified geometric framework in which cinema, painting, drawing, swipe-gesture input, and visual perception are instances of a single underlying structure: directed motion through a structured spatial or perceptual field. A film is modeled as a continuous path through a *scenario manifold* $\mathcal{S}$ whose coordinates encode camera pose, actor configuration, lighting, and environmental state. The perceptual impact of that path is determined not by its geometry in $\mathcal{S}$ but by its image under a rendering-and-perception map $\Pi : \mathcal{S} \rightarrow \mathcal{P}$ into a *perceptual manifold* $\mathcal{P}$ whose coordinates are attentional focus, revealed narrative information, occlusion structure, emotional intensity, and compositional balance. We show that cinematographic naturalness corresponds to approximate geodesic motion in $\mathcal{P}$ with respect to a metric tensor derived from information geometry, and that narrative intent introduces a potential function that forces the path toward perceptually salient states. This variational principle unifies classical editing rules (the 180-degree rule, match cuts, shot-reverse-shot) as geometric constraints on paths in $\mathcal{P}$. We then extend the framework to cover painting strokes guided by image-gradient fields, children’s drawing as a sequential grammar of spatial occupation, QWERTY swipe gestures as a trajectory language over a discrete keyboard manifold, and eye saccades as perceptual sampling paths. Across all these cases, embodied motion through a structured field generates symbolic or aesthetic meaning through the same variational mechanism. The framework connects cinematography, game engine simulation, information geometry, embodied cognition, and trajectory-based generative models.

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1 Introduction

Many cultural activities that seem superficially unrelated share a common deep structure: they are all instances of directed motion through a structured spatial or perceptual field, where meaning or aesthetic quality emerges from the shape of that motion rather than from its endpoints alone. A filmmaker choosing a camera angle, a painter laying down a brushstroke, a child completing a figure on a page, a typist swiping a word, and a viewer’s eye jumping between salient features of a scene are all performing versions of the same cognitive and bodily operation.

This paper proposes a unified mathematical framework for these activities. The central object is a *trajectory*: a continuous path $\gamma(t)$ through some structured space, where the space is endowed with a geometry that determines which paths are natural, smooth, or costly. We argue that the appropriate notion of naturalness is always determined by a *perceptual metric*—a metric tensor on a manifold of experiential states—rather than by a purely physical or symbolic metric.

The framework has five main components.

- (1) **Scenario space $\mathcal{S}$** : the high-dimensional space of all possible world states, including camera parameters, actor configurations, lighting, and environment.
- (2) **Perceptual manifold $\mathcal{P}$** : the space of experiential states as induced in a viewer, carrying a metric derived from information geometry.
- (3) **Rendering-and-perception map $\Pi : \mathcal{S} \rightarrow \mathcal{P}$** : the operator that converts scenario states into perceptual states.
- (4) **Geodesic principle**: cinematographic naturalness corresponds to approximate geodesic motion in $\mathcal{P}$ subject to a narrative potential function.
- (5) **Generality**: the same variational structure governs painting strokes, drawing grammars, swipe gestures, and eye saccades.

Modern film production already instantiates part of this structure. Virtual production systems such as Unreal Engine represent a scene as a game state $g = (C, A, L, E)$ whose components are camera pose, actor configurations, lighting, and environment, and the camera path through that state space is literally a trajectory. But a film recorded with a physical camera also constitutes a path through a latent scenario space, whether or not that space is ever explicitly simulated. The theoretical reinterpretation proposed here treats this implicit structure as the object of analysis.

Section 2 defines scenario space formally. Section 3 introduces the perceptual manifold and its metric. Section 4 states the geodesic principle and derives the variational equations. Section 5 reinterprets classical editing conventions. Section 6 grounds the metric in information geometry. Section 7 extends the framework

to painting and visual gradients. Section 8 addresses children’s drawing and the grammar of spatial action. Section 9 treats QWERTY swipe gestures as a trajectory language. Section 10 connects the framework to eye saccades and embodied cognition. Section 12 discusses implications for generative models. Section 13 concludes.

2 Scenario Space

Definition 2.1 (Scenario state). A *scenario state* is a tuple

$$s = (C, A, L, E)$$

where:

- $C = (x, y, z, \theta, \phi, f) \in \mathbb{R}^6$ encodes camera position, orientation angles, and focal length;
- $A = (a_1, \dots, a_k)$ encodes the pose and animation state of each of k actors in the scene;
- L encodes lighting configuration (source positions, intensities, color temperatures);
- E encodes environmental parameters (weather, time of day, set geometry).

The *scenario manifold* $\mathcal{S}$ is the space of all valid scenario states. For continuous physical parameters it is a smooth manifold; for discrete categorical parameters (shot type, costume) it can be replaced by a stratified space. We work in the smooth case throughout.

A *film* in this framework is a smooth path

$$\gamma : [0, T] \rightarrow \mathcal{S}.$$

Each frame corresponds to a sample $s_t = \gamma(t)$. The observable image at time t is the output of a rendering operator:

$$I_t = R(s_t),$$

where $R : \mathcal{S} \rightarrow \mathcal{I}$ maps scenario states to images in some image space $\mathcal{I}$. The observable film is therefore the sequence $(I_t)_{t \in [0, T]}$, i.e., the image of the trajectory under R .

Remark 2.2. In a real-time game engine, $\mathcal{S}$ is the engine’s world state and R is the rendering pipeline. A film recorded on a physical set has the same abstract structure, with R replaced by the physical optics of the camera. The theoretical framework applies to both.

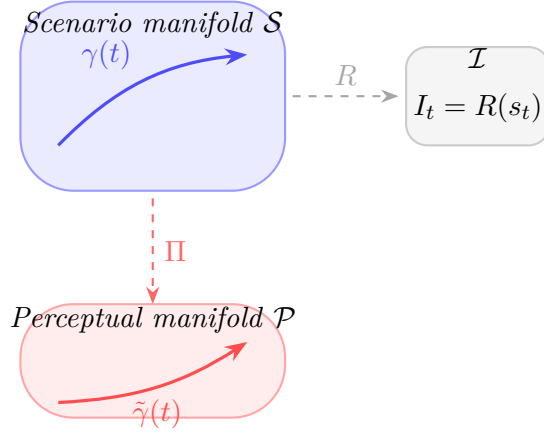


Figure 1: The scenario manifold $\mathcal{S}$, rendering map R , and perceptual manifold $\mathcal{P}$. A film is a path $\gamma(t)$ in $\mathcal{S}$; its aesthetic quality is determined by the induced path $\tilde{\gamma}(t)$ in $\mathcal{P}$.

3 The Perceptual Manifold

The scenario manifold describes what the world *is*; the perceptual manifold describes what a viewer *experiences*. A physically simple camera move can feel disorienting, while a physically elaborate tracking shot can feel effortless. The difference lies not in $\mathcal{S}$ but in $\mathcal{P}$.

Definition 3.1 (Perceptual state). A *perceptual state* is a tuple

$$p = (a, r, o, e, q) \in \mathcal{P}$$

where:

- $a \in [0, 1]$ is attentional focus (concentration of viewer attention on a primary subject);
- $r \in [0, 1]$ is revealed narrative information (cumulative proportion of withheld story information disclosed);
- $o \in [0, 1]$ is occlusion coherence (stability of depth ordering and figure-ground relations);
- $e \in [0, 1]$ is emotional intensity (arousal level induced by the scene);
- $q \in [0, 1]$ is compositional balance (degree of visual equilibrium in the frame).

The perceptual manifold $\mathcal{P}$ carries a Riemannian metric tensor $g_{ij}(p)$ that measures the cognitive cost of moving between perceptual states. The infinitesimal perceptual distance is

$$ds^2 = \sum_{i,j=1}^5 g_{ij}(p) dp^i dp^j.$$

Definition 3.2 (Rendering-and-perception map). The *rendering-and-perception*

map is a smooth map

$$\Pi : \mathcal{S} \rightarrow \mathcal{P}$$

that assigns to each scenario state the perceptual state it induces in a standard viewer. A film trajectory $\gamma(t)$ in $\mathcal{S}$ induces a *perceptual trajectory*

$$\tilde{\gamma}(t) = \Pi(\gamma(t)) \in \mathcal{P}.$$

The derivative of Π at a point $s \in \mathcal{S}$ is the Jacobian $D\Pi_s : T_s\mathcal{S} \rightarrow T_{\Pi(s)}\mathcal{P}$. The metric on $\mathcal{P}$ pulls back through Π to a metric on $\mathcal{S}$:

$$(D\Pi^*g)_s(u, v) = g_{\Pi(s)}(D\Pi_s \cdot u, D\Pi_s \cdot v), \quad u, v \in T_s\mathcal{S}.$$

In this pullback metric, directions in $\mathcal{S}$ that induce large perceptual change are far apart even if they are physically close.

4 Naturalness as a Geodesic Principle

4.1 The action functional

The perceptual length of a trajectory $\tilde{\gamma}$ over the interval $[0, T]$ is

$$L[\tilde{\gamma}] = \int_0^T \sqrt{g_{ij}(\tilde{\gamma}(t)) \dot{\tilde{\gamma}}^i(t) \dot{\tilde{\gamma}}^j(t)} dt.$$

Minimizing L is equivalent (up to reparametrization) to minimizing the *perceptual action*

$$\mathcal{A}[\tilde{\gamma}] = \frac{1}{2} \int_0^T g_{ij}(\tilde{\gamma}(t)) \dot{\tilde{\gamma}}^i(t) \dot{\tilde{\gamma}}^j(t) dt.$$

Definition 4.1 (Cinematographically natural trajectory). A perceptual trajectory $\tilde{\gamma}$ is *cinematographically natural* if it is a geodesic of $(\mathcal{P}, g)$, i.e., if it satisfies the Euler-Lagrange equations of $\mathcal{A}$.

The Euler-Lagrange equations of $\mathcal{A}$ yield the *geodesic equation*

$$\ddot{\tilde{\gamma}}^k + \Gamma_{ij}^k \dot{\tilde{\gamma}}^i \dot{\tilde{\gamma}}^j = 0,$$

where Γ_{ij}^k are the Christoffel symbols of the metric g :

$$\Gamma_{ij}^k = \frac{1}{2} g^{k\ell} (\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}).$$

The geodesic equation says that a natural camera path has no *perceptual acceleration*: it does not jerk the viewer's attention in unexpected directions.

4.2 Narrative forcing

A purely geodesic path in $\mathcal{P}$ does not capture the director's intention to reveal information or guide attention toward specific story events. We introduce a *narrative potential* $V : \mathcal{P} \rightarrow \mathbb{R}$, where low values of V correspond to perceptually important or informationally rich states. The forced trajectory satisfies

$$\frac{D\dot{\tilde{\gamma}}}{dt} = -\nabla V(\tilde{\gamma}),$$

where D/dt denotes the covariant derivative along the path. This is the geodesic equation with a forcing term: the director steers perception toward narrative attractors (a face, a weapon, an emotionally decisive gesture) at the cost of some perceptual smoothness.

Definition 4.2 (Cinematic action with narrative potential). The *cinematic action* is

$$\mathcal{J}[\tilde{\gamma}] = \int_0^T \left[\frac{1}{2} g_{ij}(\tilde{\gamma}) \dot{\tilde{\gamma}}^i \dot{\tilde{\gamma}}^j + \lambda V(\tilde{\gamma}) \right] dt,$$

where $\lambda > 0$ is a weight balancing perceptual smoothness against narrative urgency.

Proposition 4.3. The Euler-Lagrange equations of $\mathcal{J}$ are

$$\ddot{\tilde{\gamma}}^k + \Gamma_{ij}^k \dot{\tilde{\gamma}}^i \dot{\tilde{\gamma}}^j + \lambda g^{k\ell} \partial_\ell V(\tilde{\gamma}) = 0.$$

Proof. Standard variational calculus applied to the Lagrangian $\mathcal{L}(p, \dot{p}) = \frac{1}{2} g_{ij} \dot{p}^i \dot{p}^j + \lambda V(p)$. The first term gives the Christoffel-symbol term as in the pure geodesic case; the second gives the gradient of V raised by the inverse metric. $\square$

Remark 4.4. The parameter λ encodes directorial style. A director who prizes compositional fluidity (Tarkovsky) operates at small λ ; a director who prioritizes narrative information density (Eisenstein) operates at large λ .

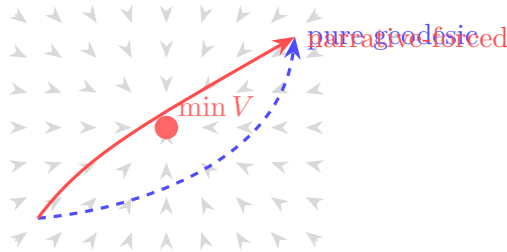


Figure 2: Narrative forcing bends the geodesic toward the attractor $\min V$ (the narratively important perceptual state) at the cost of some smoothness. The pure geodesic ignores narrative structure; the forced path balances perceptual economy with directorial intent.

5 Editing as Trajectory Transformation

Film editing introduces discontinuities or structured transformations in the scenario trajectory γ . We show how classical editing operations correspond to geometric operations on paths in $\mathcal{S}$ and $\mathcal{P}$.

5.1 Cuts

A *hard cut* at time t_0 is a jump discontinuity:

$$\gamma(t_0^-) \neq \gamma(t_0^+).$$

In $\mathcal{P}$, the corresponding perceptual jump $\tilde{\gamma}(t_0^-) \neq \tilde{\gamma}(t_0^+)$ has a perceptual cost proportional to $d_{\mathcal{P}}(\tilde{\gamma}(t_0^-), \tilde{\gamma}(t_0^+))$.

A cut is *invisible* when the perceptual distance across the cut is small, i.e., when the cut preserves attentional focus, occlusion coherence, and compositional balance. This is the geometric content of the classical principle of *continuity editing*.

5.2 Dissolves and wipes

A *dissolve* interpolates between two scenario states:

$$I_t = (1 - \alpha(t)) R(\gamma_1(t)) + \alpha(t) R(\gamma_2(t)), \quad \alpha : [0, 1] \rightarrow [0, 1].$$

In $\mathcal{P}$, this corresponds to a blended perceptual path. The dissolve is perceptually smooth when the two source paths $\Pi(\gamma_1)$ and $\Pi(\gamma_2)$ are close in $\mathcal{P}$.

5.3 Montage

A *montage sequence* is a piecewise trajectory $\gamma = (\gamma_1, \gamma_2, \dots, \gamma_k)$ with jump discontinuities at the joints. In contrast to continuity editing, montage deliberately creates large perceptual jumps. The Eisenstein principle is that conceptually or emotionally significant jumps along certain coordinates of $\mathcal{P}$ (especially e and r) can generate meaning even at high perceptual cost.

5.4 The 180-degree rule

The 180-degree rule constrains the camera trajectory to one side of an imaginary line connecting the principal actors. In scenario-space terms, if actors define an axis $\ell \subset \mathbb{R}^3$, the camera must satisfy

$$C \in \mathcal{H}_+ = \{C : \langle n, \pi_\ell(C) \rangle > 0\}$$

for a normal n to ℓ . Crossing this line corresponds to a large jump in the occlusion coordinate o of $\mathcal{P}$, because the left-right mapping of actors is reversed. The 180-degree rule thus minimizes perceptual cost in the o coordinate; violating it forces a large excursion in $\mathcal{P}$.

Proposition 5.1. *The 180-degree rule is the condition that the camera trajectory remains within a single chart of $\mathcal{P}$ with respect to the occlusion coordinate o . Crossing the line corresponds to a coordinate singularity in that chart.*

5.5 Match cuts and eyeline matches

A *match cut* transitions between two shots whose perceptual states are closely aligned: similar attentional focus a , similar compositional balance q , and continuous motion energy in e . Formally, a match cut at time t_0 satisfies

$$\|\tilde{\gamma}_1(t_0^-) - \tilde{\gamma}_2(t_0^+)\|_{\mathcal{P}} < \varepsilon$$

for small ε . The match cut is the editorial operation that minimizes the perceptual distance across a cut.

An *eyeline match* aligns the attentional direction of one shot with the implied object of attention in the next, preserving continuity in the a coordinate of $\mathcal{P}$.

6 Information Geometry of the Perceptual Metric

6.1 Viewer attention as a probability distribution

We ground the metric g on $\mathcal{P}$ in information geometry by modeling the viewer’s attentional state as a probability distribution over scene elements.

Let $\mathcal{X}$ be the set of salient objects in the frame (faces, hands, weapons, etc.). The viewer’s attentional state at time t is a distribution

$$P_t \in \Delta(\mathcal{X}),$$

the probability simplex over $\mathcal{X}$. The *attentional cost* of moving from P_t to P_{t+dt} is measured by a divergence in this simplex.

6.2 The Fisher-Rao metric

When P_t is parameterized by coordinates $\theta \in \Theta$, the *Fisher information metric* is

$$g_{ij}^F(\theta) = \mathbb{E}_{P(\cdot; \theta)} \left[\frac{\partial \log P(x; \theta)}{\partial \theta^i} \frac{\partial \log P(x; \theta)}{\partial \theta^j} \right].$$

This is the natural Riemannian metric on the statistical manifold $\{P(\cdot; \theta) : \theta \in \Theta\}$. Geodesics in this metric correspond to paths of least inferential surprise: the viewer’s belief about what is important in the frame changes as smoothly as possible.

Theorem 6.1. *The perceptual metric g on $\mathcal{P}$ can be taken to be (or to dominate) the Fisher-Rao metric on the manifold of attentional distributions. A camera trajectory is perceptually smooth if and only if the induced path in $\Delta(\mathcal{X})$ has bounded Fisher-Rao velocity.*

Proof sketch. The attentional coordinate a of $\mathcal{P}$ determines the centroid of the distribution P_t over salient objects. The other coordinates (r, o, e, q) introduce additional geometric structure. Since the Fisher-Rao metric is the unique (up to scale) Riemannian metric on $\Delta(\mathcal{X})$ that is invariant under sufficient statistics [3], bounding its velocity bounds the rate of change of all attentional quantities. The claim follows from the smoothness of the parameterization by (a, r, o, e, q) . $\square$

6.3 Continuity editing as minimal attentional transport

Continuity editing works by keeping the attentional transport cost low across cuts. Formally, a cut at time t_0 transitions from distribution P^- to distribution P^+ . The cost is

$$W_2(P^-, P^+) = \inf_{\pi \in \Pi(P^-, P^+)} \left(\int d(x, y)^2 d\pi(x, y) \right)^{1/2},$$

where W_2 is the Wasserstein-2 distance and the infimum is over couplings π . A cut is invisible when $W_2(P^-, P^+)$ is small: the viewer’s attention can be transported cheaply from the pre-cut to the post-cut distribution.

This gives a precise measure-theoretic account of why continuity editing works: it minimizes the optimal transport cost of viewer attention across shot boundaries.

7 Painting as Gradient-Field Integration

7.1 The image as a scalar field

A painting-in-progress at time t can be represented as a scalar field $B_t : \Omega \rightarrow \mathbb{R}$, where $\Omega \subset \mathbb{R}^2$ is the canvas and $B_t(x, y)$ encodes luminance or tonal value. A brushstroke is a trajectory

$$\gamma_{\text{paint}} : [0, \tau] \rightarrow \Omega$$

that updates the canvas field:

$$B_{t+dt}(x, y) = B_t(x, y) + K(x - \gamma_x(t), y - \gamma_y(t)),$$

where K is the brush footprint kernel. Painting is therefore the process of *writing trajectories into a field*.

7.2 The gradient field and isophotes

The structural information in the image is encoded in the gradient field

$$\nabla B = \left(\frac{\partial B}{\partial x}, \frac{\partial B}{\partial y} \right).$$

In computer vision this is approximated by Sobel filters:

$$G_x = S_x * B, \quad G_y = S_y * B, \quad |\nabla B| = \sqrt{G_x^2 + G_y^2}.$$

Painters often orient strokes along the *isophote* direction, tangent to level curves of B :

$$T = (-B_y, B_x).$$

A stroke guided by this field satisfies

$$\frac{d\gamma_{\text{paint}}}{dt} = \alpha T(\gamma_{\text{paint}}(t)) + \beta \nabla B(\gamma_{\text{paint}}(t)),$$

where α controls how much the stroke follows the form contour and β controls how much it follows the tonal gradient.

7.3 Structure tensor and stroke direction

The local orientation of image structure can be summarized by the *structure tensor*

$$J = \begin{pmatrix} B_x^2 & B_x B_y \\ B_x B_y & B_y^2 \end{pmatrix}.$$

The dominant eigenvector of J gives the preferred local orientation of edges and forms. A painter's stroke tends to follow this direction because it preserves the perceived structure of the surface.

Proposition 7.1. *A painter who aligns brushstrokes with the dominant eigenvector field of J is integrating a vector field derived from the tonal gradient of the scene. Each stroke is a geodesic segment in the metric induced by J on Ω .*

7.4 Comparison with cinematography

The analogy with the cinematic case is precise. In both systems:

- there is a trajectory $\gamma(t)$ through a structured space;

- there is a field (tonal gradient in Ω ; perceptual metric on $\mathcal{P}$) that determines which paths are natural;
- naturalness corresponds to geodesic or near-geodesic motion in that field.

The difference is that in painting the trajectory modifies the field it moves through (the canvas changes with each stroke), while in cinematography the trajectory merely samples a pre-existing scenario manifold.

8 Drawing as a Sequential Grammar of Spatial Action

8.1 Goodnow’s grammar of action

Jacqueline Goodnow’s empirical studies of children’s drawing demonstrate that drawings are not arbitrary marks but records of structured action sequences [6]. Children follow systematic strategies for initiating figures, attaching new elements, and expanding outward. Goodnow called these strategies a *grammar of action*: a set of procedural rules governing how actions combine to produce spatial structures.

In the framework developed here, a drawing is a sequence of trajectories

$$\gamma_1, \gamma_2, \dots, \gamma_n$$

each depositing a stroke on the canvas field B . The grammar of action governs the conditional distribution of γ_{k+1} given the current canvas state B_{t_k} after the k -th stroke.

Formally, let the drawing process be an iterative mapping

$$B_{t+1} = F(B_t, a_t),$$

where a_t is the stroke action chosen at step t according to the grammar. Goodnow’s finding that children follow systematic attachment and expansion rules means that F is not arbitrary: new strokes are placed near existing structures, extend boundaries, or occupy regions that preserve compositional balance.

8.2 Developmental stages as metric learning

Goodnow’s developmental progression from simple attachment rules in young children to symmetric, hierarchically organized compositions in older children can be interpreted as the progressive acquisition of a more refined metric on the space of compositional states. Young children use a coarse metric in which proximity is the dominant criterion. Older children use a metric that also weighs symmetry, distribution, and relational coherence.

8.3 Kellogg’s primitive trajectories

Rhoda Kellogg’s catalog of recurring scribble forms in children’s early drawing provides a vocabulary of trajectory primitives [7]:

- loops (near-circular closed paths);
- spirals (outwardly expanding loops);
- zigzags (alternating diagonal segments);
- arches (smooth convex arcs);
- radiating lines (spokes from a central point).

These primitives correspond to the simplest geodesic shapes available in the plane: constant-curvature curves (circles, arches), piecewise-linear paths (zigzags), and self-similar expanding paths (spirals). Kellogg’s observation that these forms appear cross-culturally before representational drawing develops supports the claim that the trajectory grammar is prior to, and independent of, semantic representation.

9 QWERTY Swipe as a Trajectory Language

9.1 The keyboard as an embedded graph

Let $K = (V, E)$ be the QWERTY keyboard graph, where each key is a vertex $v \in V$ and edges connect neighboring keys. The graph is embedded in the plane by the coordinate map

$$\phi : V \rightarrow \mathbb{R}^2,$$

which assigns each key its physical center position. This embedding makes K a *finite metric space* in the induced Euclidean metric.

Definition 9.1 (Swipe word trajectory). For a word $w = w_1w_2 \cdots w_n$ over the keyboard alphabet $\Sigma = V$, the *swipe trajectory* is the piecewise-linear path

$$P(w) = (\phi(w_1), \phi(w_2), \dots, \phi(w_n)) \subset \mathbb{R}^2.$$

Theorem 9.2 (Trajectory encoding). *The map $P : \Sigma^* \rightarrow \mathcal{T}$ from words to piecewise-linear trajectories is a compositional encoding of the free monoid $(\Sigma^*, \cdot)$ into a space $\mathcal{T}$ of planar paths. Hence QWERTY swipe induces a formal language for representing trajectory classes.*

Proof. The concatenation $w = uv$ of two words maps to the path concatenation $P(w) = P(u) \cdot P(v)$ (joined at the endpoint of $P(u)$). This is the monoid homomorphism property. The map is nontrivial because distinct words generically produce distinct polygonal chains. Hence P is a nontrivial homomorphism from $(\Sigma^*, \cdot)$ into $(\mathcal{T}, \cdot)$. $\square$

9.2 Geometric invariants of words

Each word w has a geometric signature under P :

$$\begin{aligned}
L(w) &= \sum_{i=1}^{n-1} \|\phi(w_{i+1}) - \phi(w_i)\| && \text{(path length),} \\
C(w) &= \sum_{i=2}^{n-1} |\theta_i| && \text{(turn complexity),} \\
X(w) &= (\text{number of self-intersections of } P(w)), \\
\kappa_{\text{close}}(w) &= 1 - \frac{\|\phi(w_n) - \phi(w_1)\|}{L(w)} && \text{(closure index).}
\end{aligned}$$

Definition 9.3 (Trajectory equivalence). Two words $w, u \in \Sigma^*$ are *trajectory-equivalent* if their normalized swipe paths are within ε in Fréchet distance:

$$w \sim u \iff d_F(\tilde{P}(w), \tilde{P}(u)) < \varepsilon,$$

where $\tilde{P}$ denotes the arc-length-normalized version of P .

Trajectory equivalence classes define a *gesture morphology*: families of words that share a common kinematic shape regardless of their semantic or phonological properties.

9.3 Examples: geometric taxonomy of words

Informal inspection of QWERTY swipe trajectories reveals that common words fall into recognizable shape families:

Shape class	Example word	Dominant feature
Loop / circle	<i>tent</i>	high closure index κ_{close}
Rectilinear	<i>route</i>	few turn directions, dominant orthogonal phases
Braided / pretzel	<i>magic</i>	$X(w) \geq 1$ (self-intersection)
Arch	<i>con</i>	smooth convex arc, $C(w)$ small
Hook	<i>do</i>	short path, large single turn
Ribbon sweep	<i>all</i>	large $L(w)$, small $C(w)$, near-linear
Spiral	<i>crying</i>	increasing radius, monotone turning
Zigzag	<i>Zeno</i>	alternating diagonal segments

Corollary 9.4. *Any lexicon $\mathcal{L} \subset \Sigma^*$ inherits a trajectory semantics via P , making the lexicon simultaneously a symbolic language and a geometric codebook. The natural language vocabulary is a proper submanifold of the full swipe trajectory space.*

9.4 The keyboard as a discrete canvas

The keyboard can be interpreted as a discretized version of the painter’s canvas: instead of depositing pigment continuously, the gesture snaps to discrete letter centers. Swipe typing is therefore a quantized painting system in which the interpretation of a trajectory produces symbols rather than pigment.

The dual reading of a swipe word is:

$$w \mapsto (\text{semantic meaning}, \text{gesture class}).$$

Natural language words function as compressed geometric operators: each word is simultaneously semantic and kinematic.

10 Eye Saccades and Embodied Perception

10.1 Saccadic paths as perceptual trajectories

Visual perception does not occur through smooth continuous scanning. The eye moves through sequences of rapid fixation jumps called *saccades*. If fixation points are $p_1, p_2, \dots, p_n \in \mathcal{I}$, the visual scan path is a trajectory

$$\gamma_{eye}(t)$$

through the image plane $\mathcal{I}$.

Saccades are not random: they are guided by a *saliency field* $\sigma : \mathcal{I} \rightarrow \mathbb{R}_{\geq 0}$, where high saliency corresponds to edges, faces, motion, and high-contrast regions. The saccade trajectory approximately maximizes accumulated saliency subject to a motor cost for large eye movements. This is a constrained optimization structurally identical to the variational principle of Section 4.

10.2 Three trajectory systems in the perceptual-motor loop

We now see that three trajectory systems operate jointly in visual culture:

System	Body part	Output
Eye saccades	Ocular muscles	Perceptual sampling
Brush strokes	Arm and hand	Image construction
Swipe gestures	Finger	Symbol generation
All three: trajectory $\gamma(t)$ through a structured field		

The hand trajectory of the painter tends to follow the same image features that the eye’s saccades follow (edges, contours, saliency peaks). This is because both are

guided by the same gradient field of the image: the painter is externalizing the scan path of the visual system as a stroke.

10.3 Mimetic hypothesis and motor simulation

Arnie Cox’s *mimetic hypothesis* holds that music perception recruits internal motor representations: listeners simulate the bodily actions that could produce the perceived sounds [4]. The perceptual trajectory in that case moves through *motor activation space* rather than image space.

Central pattern generators (CPGs) provide a neural substrate for such simulations. A CPG chain produces rhythmic sequences of motor states:

$$m_{t+1} = F(m_t),$$

a trajectory through motor state space that can be driven by auditory input. Baddeley’s phonological loop is a cyclic CPG:

$$a_1 \rightarrow a_2 \rightarrow \cdots \rightarrow a_n \rightarrow a_1,$$

maintaining verbal information through repeated traversal of an articulatory trajectory.

The unified claim is therefore:

Perception and expression across modalities share a common structure: they are trajectories through structured fields, whether those fields are spatial gradients on a canvas, geometric lattices on a keyboard, saliency maps in an image, or motor activation manifolds in the nervous system.

11 Developmental Arc: From Object Grammar to Gradient Field

Artistic development can be interpreted as a shift in the representational model underlying trajectory production. This shift moves through three conceptual layers.

Stage 1: Object-oriented. The page is treated as a set of discrete entities. A drawing is an object graph $G = (R, E)$ where regions R_i correspond to symbolic shapes (circle for head, lines for limbs) and edges E encode attachment or containment. Strokes follow object boundaries. This is the stage described by Goodnow’s grammar of action in early childhood.

Stage 2: Field-based. The page becomes a tonal field $I(x, y)$. Strokes follow the gradient and isophote structure of the field rather than symbolic object boundaries.

The artist observes continuous phenomena (shadow, reflection, graduated tone) and reconstructs them as trajectories through the gradient field.

Stage 3: Radiance model. The experienced artist understands the image as the projection of a physical light-transport process. Strokes implicitly reconstruct the radiance field, solving the inverse problem:

$$L_o(x, \omega) = L_e(x, \omega) + \int_{\Omega} f_r(x, \omega', \omega) L_i(x, \omega') (\omega' \cdot n) d\omega'.$$

The artist does not compute this integral, but has internalized how light behaves and places strokes that reproduce its effects. In computer graphics this is the rendering equation; in artistic practice it is the implicit model that separates expert from novice draughtsmen.

This three-stage arc corresponds to a progression from a symbolic object metric to an information-geometric field metric as the criterion for naturalness of strokes.

12 Learning Cinematic Trajectories

The variational framework suggests a natural approach to learning cinematic style from observed films.

Given a corpus of films with associated perceptual state sequences $\tilde{\gamma}_1, \tilde{\gamma}_2, \dots, \tilde{\gamma}_N$ (approximated through automatic annotation of attention, composition, and narrative labels), one can learn:

- (a) the metric g_θ on $\mathcal{P}$ by fitting a Riemannian metric to minimize geodesic deviation of observed trajectories;
- (b) the potential V_ϕ by fitting the forced geodesic equation to observed perceptual paths;
- (c) the rendering-and-perception map Π_ψ by learning to predict perceptual coordinates from scenario states.

A generative cinematography system would then sample new scenario trajectories $\gamma(t)$ that induce target perceptual trajectories $\tilde{\gamma}(t)$ minimizing $\mathcal{J}[\tilde{\gamma}]$ for a specified narrative potential.

Remark 12.1. This framework is compatible with diffusion-based video generation models, where the denoising trajectory can be interpreted as navigation through scenario space. Conditioning on a perceptual metric and narrative potential constrains the generative trajectory to cinematographically natural paths.

Applications include:

- automatic cinematography for virtual production environments;

- camera planning algorithms in game engines;
- film style analysis and director fingerprinting;
- trajectory-based video generation with aesthetic constraints;
- gesture-driven interfaces for creative tools.

13 Conclusion

We have proposed a unified geometric framework in which diverse cultural practices—cinema, painting, drawing, swipe-gesture input, and visual perception—share a common mathematical structure: directed motion through a structured field, where naturalness is determined by a variational principle on a Riemannian manifold.

The central theoretical contribution is the *geodesic principle of cinematography*: natural camera movement approximates a geodesic in the perceptual manifold $\mathcal{P}$, subject to forcing by a narrative potential V . This principle unifies classical editing conventions (continuity, the 180-degree rule, match cuts, montage) as geometric constraints on paths in $\mathcal{P}$, and grounds them in information geometry through the Fisher-Rao metric on the manifold of viewer attentional distributions.

The same variational structure governs the painter’s brush (guided by tonal gradient fields), the child’s drawing hand (constrained by a grammar of spatial action), the typist’s swiping finger (moving through a discrete keyboard manifold), and the viewer’s eye (jumping between salient image features). Across all these modalities, meaning and aesthetic quality emerge from the shape of motion through a structured field, not from the endpoint of that motion.

This framework has both theoretical and practical implications. Theoretically, it provides a unifying vocabulary for disparate fields: cinematography, embodied cognition, information geometry, art education, and human-computer interaction. Practically, it suggests principled methods for learning and generating cinematographically natural trajectories, designing gesture interfaces whose vocabulary is geometrically coherent, and analyzing artistic style as a characteristic signature in trajectory space.

The deeper implication is that human creative and perceptual culture is, to a significant degree, a civilization of gesture: the organized motion of bodies through structured fields, leaving traces that carry meaning because of the geometry of how they were made.

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